UNIT IV

Chapter 11

Chemical Coordination and Integration

Chapter Outline

11.1 Endocrine glands and hormones
11.2 Human endocrine system
11.3 Hypo and hyper activity of endocrine glands and related disorders
11.4 Mechanism of hormone action.



Klotho an anti-aging hormone makes people smart, enhances cognitive abilities and longevity.

Learning Objectives:

- *Understands the positions of the various endocrine glands and their secretions.*
- *Learns the mechanism of hormone action.*
- *Understands the disorders related to hypo and hyper activity of the endocrine glands.*
- *Learns the role of gastro intestinal hormones.*

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While hearing your test marks, some may have anxiety and some may hesitate to hear and some may be worried. Do you know the reasons for such immediate changes? While seeing any unexpected happenings, we get goose bumps. Do you know the reason, why?

These are all due to the biochemical changes happening in our body, Which are created by the endocrine system. The above mentioned biochemical changes are due to the hormone adrenalin (Epinephrine or flight, fright and fight hormone).

11.1 Endocrine glands and hormones

Physiological functions of our human body is regulated and coordinated by both neural and endocrine systems. The endocrine system influences the metabolic activities by means of **hormones** (hormone means *to excite*) which are chemical messengers released into the blood and circulated as chemical signals and acts specifically on certain organs or tissues called target organs or target tissues. Hormones may speed up or slow down or alter the activity of the target organs. The hormones secreted do not remain permanently in the blood but are converted by the liver into inactive compounds and excreted by the kidneys.

Hormones are chemical messengers because they act as organic catalysts and coenzymes to perform specific functions in the target organs. The target organs contain receptor molecules either on the surface or within the cell. Although different hormones come in contact with cells, only

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